

Terna Engineering College
Computer Engineering Department

Class: TE

Sem.: VI

Course: System Security Lab

PART A

(PART A : TO BE REFFERED BY STUDENTS)

Experiment No.06

A.1 Aim: Design a network and implement packet sniffing on telnet traffic using wireshark.

A.2 Prerequisite:

1. Basic Knowledge of IP addresses, Port numbers, TCP and UDP Protocols.

A.3 Outcome:

After successful completion of this experiment students will be able to

Apply basic network command to gather basic network information.

A.4 Theory:

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color-coding and other features that let you dig deep into network traffic and inspect individual packets.

Features of Wireshark :

- Available for UNIX and Windows.
- Capture live packet data from a network interface.
- Open files containing packet data captured with tcpdump/WinDump, Wireshark, and a number of other packet capture programs.
- Import packets from text files containing hex dumps of packet data.
- Display packets with very detailed protocol information.

- Export some or all packets in a number of capture file formats.
- Filter packets on many criteria.
- Search for packets on many criteria.

SNO .	NAME OF THE DEVICE	INTERFA CE	IP ADDRESS	Subnet Mask	Default Gateway
1.	Router 0	g0/0	192.168.120.1	255.255.255.0	-----
2.	PC	Fast Ethernet	192.168.120.2	255.255.255.0	192.168.120.1

- Colorize packet display based on filters.
- Create various statistics.

Capturing Packets

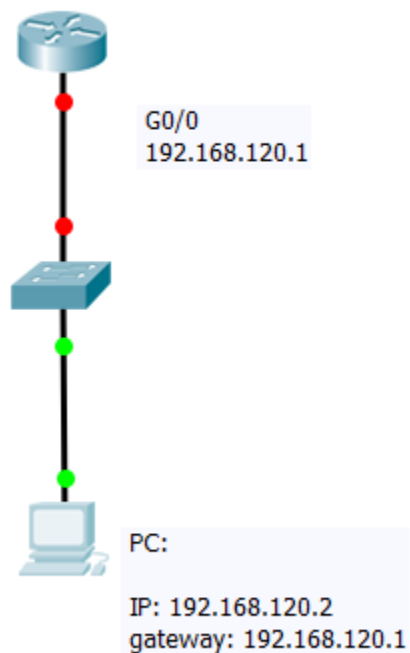
After downloading and installing wireshark, you can launch it and click the name of an interface under Interface List to start capturing packets on that interface. For example, if you want to capture traffic on the wireless network, click your wireless interface. You can configure advanced features by clicking Capture Options.

Filtering Packets

If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic. Still, you'll likely have a large amount of packets to sift through. That's where Wireshark's filters come in. The most basic way to apply a filter is by typing it into the filter box at the top of the window and clicking Apply (or pressing Enter). For example, type `—dns` and you'll see only DNS packets. When you start typing, Wireshark will help you autocomplete your filter.

A5. Interface Configuration table

A6. Design



PART B

(PART B : TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)

Roll No.C54

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Class :TE-C	Batch :C3
Date of Experiment: 27-2-25	Date of Submission: 6-3-25
Grade :	

B.1 Output

(add snapshot of output)

The screenshot displays the Cisco Packet Tracer interface. The top section shows a network topology with three devices: Router0 (labeled 'Router-PT'), Switch1 (labeled 'Switch-PT'), and PC0 (labeled 'PC-PT'). They are connected in a vertical line. The bottom section shows a packet capture window titled 'Capturing from Ethernet'. The window displays a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The packets are filtered by 'Apply a display filter: <<Ctrl+>>'. The list shows various protocols including IGMPv2, ARP, HDNS, ARP, ICMPv6, SSDP, and ARP. The bottom of the window shows the packet details for the selected packet, including the Ethernet II header and the ARP request.

No.	Time	Source	Destination	Protocol	Length	Info
10662	69.061392	192.168.6.1	224.0.0.251	IGMPv2	60	Membership Report group 224.0.0.251
10663	69.061392	HP_b8:b3:63	Broadcast	ARP	60	Who has 192.168.3.211? Tell 192.168.7.182
10664	69.063332	192.168.6.1	224.0.0.251	HDNS	124	Standard query 0x0000 PTR_companion-link_tcp.local, "QU" question PTR_rdlink_tcp.local, "QU" question PTR_...
10665	69.063332	fe80::189c:3637:40b...	ff02::fb	HDNS	144	Standard query 0x0000 PTR_companion-link_tcp.local, "QU" question PTR_rdlink_tcp.local, "QU" question PTR_...
10666	69.071975	Dell_5d:b9:d6	Broadcast	ARP	60	Who has 192.168.0.85? Tell 192.168.3.241
10667	69.079629	HewlettP_43:8f:68	Broadcast	ARP	60	Who has 192.168.0.43? Tell 192.168.1.146
10668	69.085366	HewlettP_5d:cb:35	Broadcast	ARP	60	Who has 192.168.6.238? Tell 192.168.2.156
10669	69.090407	fe80::9857:7afa:337...	ff02::1:ffdc:352b	ICMPv6	86	Neighbor Solicitation for fe80::c665:16ff:fedc:352b from f4:39:09:4a:ac:36
10670	69.101378	192.168.2.163	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
10671	69.118582	Sophos_6c:88:b6	Broadcast	ARP	60	Who has 192.168.7.85? Tell 192.168.1.254
10672	69.127598	192.168.6.148	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
10673	69.175993	Sophos_6c:88:b6	Broadcast	ARP	60	Who has 192.168.0.98? Tell 192.168.1.254
10674	69.179958	Sophos_6c:88:b6	Broadcast	ARP	60	Who has 192.168.0.103? Tell 192.168.1.254
10675	69.182895	HP_b8:84:3d	Broadcast	ARP	60	Who has 192.168.3.146? Tell 192.168.1.124
10676	69.212197	d6:f1:59:5a:ea:36	Broadcast	ARP	60	Who has 192.168.6.148? Tell 192.168.6.12
10677	69.221198	Dell_5d:c2:db	Broadcast	ARP	60	Who has 192.168.4.178? Tell 192.168.1.67

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{B8B594F-ECB7-4727-AE48-6865E5F5600}, id 0
 > Ethernet II, Src: Dell_22:2a:c9 (b8:ac:6f:22:2a:c9), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 > Address Resolution Protocol (request)

```

0000  ff ff ff ff ff ff b8 ac 6f 22 2a c9 00 06 00 01  .....o**.....
0010  00 00 06 04 00 01 b8 ac 6f 22 2a c9 c0 a8 83 e4  .....o**.....
0020  00 00 00 00 00 00 c0 a5 00 55 00 00 00 00 00 00  .....U.....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....

```

Ethernet: <live capture in progress> | Packets: 10677 | Displayed: 10677 (100.0%) | Profile: Default

Wireshark interface showing network traffic analysis. The top pane displays a list of captured packets, including DNS queries and responses. The middle pane shows the details of the selected packet (Frame 80799), highlighting the Multicast Domain Name System (query) section. The bottom pane displays the raw packet data in hexadecimal and ASCII format.

Packets: 101693 · Displayed: 6783 (6.7%)

Profile: Default

Frame 80799: 127 bytes on wire (1016 bits), 127 bytes captured (1016 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE4B-6B656E5F5680}, id 0

Ethernet II, Src: HewlettP_59:6c:2e (3c:52:82:59:6c:2e), Dst: IPv6mcast_fb (33:33:00:00:00:fb)

Internet Protocol Version 6, Src: fe80::e8cd:e77b:fe7_ ff02::fb

User Datagram Protocol, Src Port: 5353, Dst Port: 5355

Multicast Domain Name System (query)

Raw Data:

```
0000 33 33 00 00 00 fb 3c 52 82 59 6c 2e 86 dd 60 0e 33:---<R>V1...
0010 bc 58 00 49 11 01 fe 00 00 00 00 00 00 00 e3 cd X I .....
0020 e7 7b fe 7a c7 0e ff 02 00 00 00 00 00 00 00 00 -{ z .....
0030 00 00 00 00 00 00 14 e9 14 e9 00 49 2d b9 00 00 .....:.....
0040 00 00 00 01 00 00 00 00 00 00 1e 48 50 20 4c 61 .....HP La
0050 73 65 72 4a 65 74 20 50 72 6f 20 4d 37 30 36 6e serJet P ro H706n
0060 20 28 41 42 34 45 41 31 29 05 5f 69 70 70 73 04 (AB4EA1)_ipps
```

Wireshark capture of traffic on the Ethernet interface. The packet list shows several DNS queries and one ICMP destination unreachable message. The packet details pane shows the structure of the selected ICMP packet (15342), including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol fields. The packet bytes pane displays the raw hex and ASCII data.

No.	Time	Source	Destination	Protocol	Length	Info
14430	93.542282	192.168.3.177	103.14.232.100	DNS	96	Standard query 0x880e A evoke-windowservices-tas.msedge.net
14452	93.675804	192.168.3.177	103.14.232.100	DNS	77	Standard query 0xa4c2 A ccsp.digicert.com
14540	94.713195	192.168.3.177	103.14.232.100	DNS	77	Standard query 0xa4c2 A ccsp.digicert.com
14658	95.716321	192.168.3.177	103.14.232.100	DNS	77	Standard query 0xa4c2 A ccsp.digicert.com
15336	100.717164	192.168.3.177	103.14.232.100	DNS	77	Standard query 0xa4c2 A ccsp.digicert.com
15342	100.721102	192.168.3.177	103.14.232.100	ICMP	105	Destination unreachable (Port unreachable)
60377	362.990196	192.168.3.177	103.14.232.100	DNS	92	Standard query 0x432e A tiu.dl.delivery.mp.microsoft.com

Frame 15342: 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE48-6B656E5F5600}, id 0

Ethernet II, Src: Dell_cale:f2 (bc:30:5b:ca:1e:f2), Dst: Sophos_6c:88:b6 (00:1a:8c:6c:88:b6)

Internet Protocol Version 4, Src: 192.168.3.177, Dst: 103.14.232.100

Internet Control Message Protocol

Domain Name System (response)

0000 00 1a 8c 6c 88 b6 bc 30 5b ca 1e f2 00 00 45 00 ...1...0 [.....E..

0010 00 5b ec 93 00 00 00 01 00 00 c0 a8 03 b1 67 0e [...].....g..

0020 e8 64 03 03 11 06 00 00 00 00 45 70 00 3f 53 a9 [...].....F...P..

0030 00 00 38 11 1a d9 67 0e e8 64 c0 a8 03 b1 00 35 [...].....g...d.....5

0040 f3 22 00 2b e4 42 a4 c2 81 05 00 01 00 00 00 00 [...].....+B.....

0050 00 00 64 c7 63 79 00 64 69 07 09 03 65 72 74 [...].....uasp digicert

0060 03 63 6f 6d 00 00 01 00 01com.....

Wireshark capture of traffic on the Ethernet interface. The packet list shows several DNS queries and one ARP request. The packet details pane shows the structure of the selected ARP packet (54777), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol fields. The packet bytes pane displays the raw hex and ASCII data.

No.	Time	Source	Destination	Protocol	Length	Info
65259	394.210352	192.168.3.203	224.0.0.251	NDNS	107	Standard query 0x0000 SRV HP LaserJet Pro M706n (AB4EA1) _ipps._tcp.local, "QM" question
65260	394.210352	fe80::e8cd:e77b:fe7...ff02::fb	224.0.0.251	NDNS	127	Standard query 0x0000 SRV HP LaserJet Pro M706n (AB4EA1) _ipps._tcp.local, "QM" question
65261	394.210024	192.168.3.203	224.0.0.251	NDNS	107	Standard query 0x0000 TXT HP LaserJet Pro M706n (AB4EA1) _ipps._tcp.local, "QM" question
65262	394.210024	fe80::e8cd:e77b:fe7...ff02::fb	224.0.0.251	NDNS	127	Standard query 0x0000 TXT HP LaserJet Pro M706n (AB4EA1) _ipps._tcp.local, "QM" question
65263	394.212451	HewlettP_59:6c:12e	Broadcast	ARP	60	Who has 192.168.0.85? Tell 192.168.3.203
65264	394.223564	192.168.2.163	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
65265	394.238323	192.168.4.70	224.0.0.251	NDNS	75	Standard query 0x0000 A NPIDC658F.local, "QM" question
65266	394.238323	fe80::d368:1171:1f5...ff02::fb	224.0.0.251	NDNS	95	Standard query 0x0000 A NPIDC658F.local, "QM" question
65267	394.238323	192.168.4.70	224.0.0.251	NDNS	75	Standard query 0x0000 AAAA NPIDC658F.local, "QM" question
65268	394.238671	192.168.4.70	224.0.0.252	LLMNR	60	Standard query 0xa7b4 A NPIDC658F
65269	394.238675	fe80::d368:1171:1f5...ff02::fb	224.0.0.251	NDNS	95	Standard query 0x0000 AAAA NPIDC658F.local, "QM" question
65270	394.238675	fe80::d368:1171:1f5...ff02::1:3	224.0.0.252	LLMNR	89	Standard query 0xa7b4 A NPIDC658F
65271	394.239013	192.168.4.70	224.0.0.252	LLMNR	69	Standard query 0x6b37 AAAA NPIDC658F
65272	394.239015	fe80::d368:1171:1f5...ff02::1:3	224.0.0.252	LLMNR	89	Standard query 0x6b37 AAAA NPIDC658F
65273	394.267967	Dell_5d:c6:5c	Broadcast	ARP	60	Who has 192.168.0.85? Tell 192.168.1.34
65274	394.291766	HewlettP_43:fb:ab	Broadcast	ARP	60	Who has 192.168.0.85? Tell 192.168.6.43

Frame 54777: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE48-6B656E5F5600}, id 0

Ethernet II, Src: Sophos_6c:88:b6 (00:1a:8c:6c:88:b6), Dst: Dell_cale:f2 (bc:30:5b:ca:1e:f2)

Internet Protocol Version 4, Src: 210.50.203.55, Dst: 192.168.3.177

Transmission Control Protocol, Src Port: 443, Dst Port: 49880, Seq: 3128, Ack: 37723, Len: 0

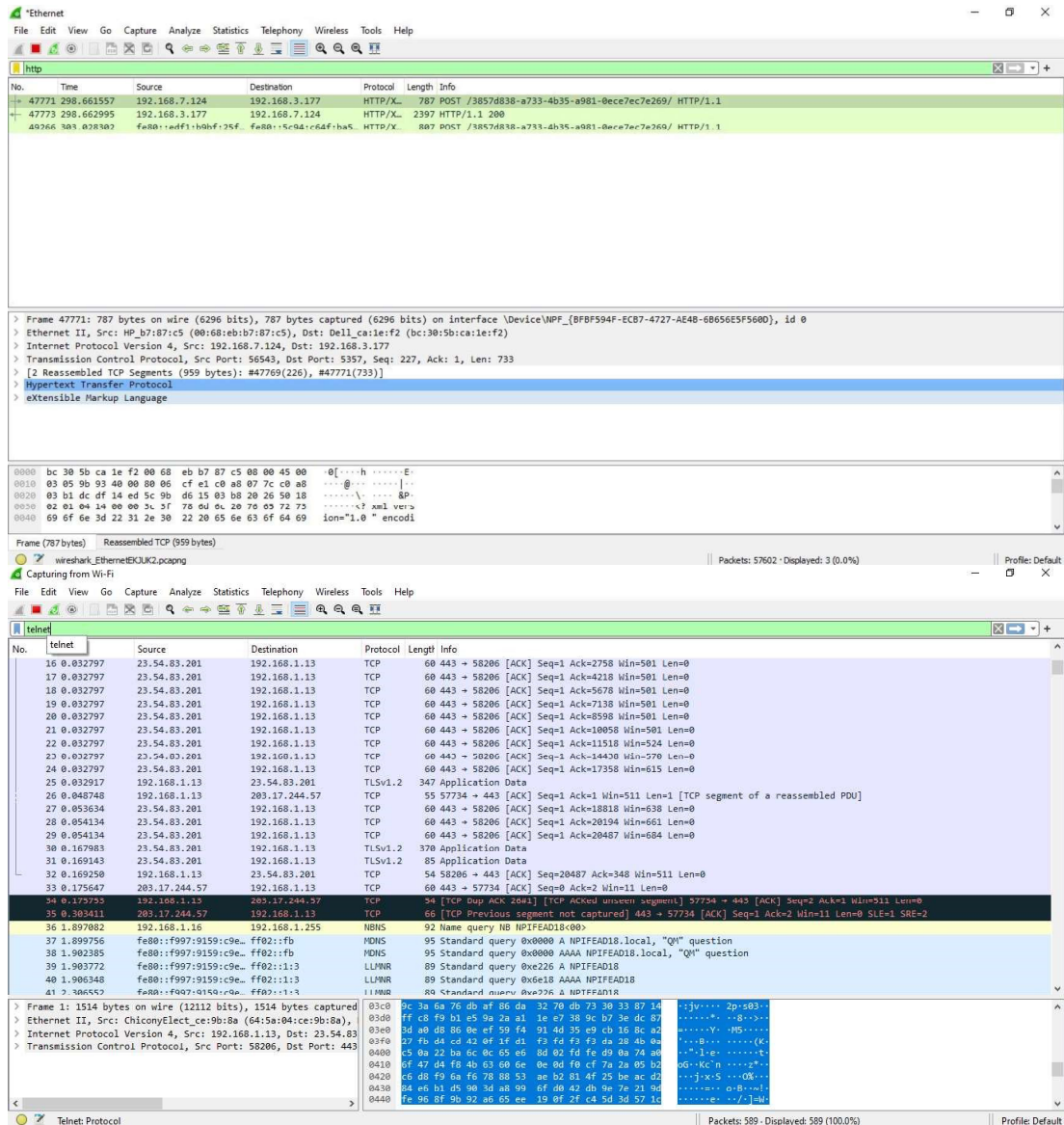
0000 bc 30 5b ca 1e f2 00 1a 8c 6c 88 b6 00 00 45 00 0[.....].....E..

0010 00 34 8b e8 00 00 7a 06 4c a4 d8 3a cb 23 c0 a8 [...].....z...L...#..

0020 03 b1 01 0b c3 3c 47 b0 b6 93 4b fb b7 3f 00 10 [...].....K...7...#..

0030 00 00 00 28 00 00 01 01 05 0a 4b fb b7 3c 4b fb [...].....K...3...#..

0040 07 11com.....



B.2 Commands / tools used with syntax:

•

Tools:

1. Wireshark: A packet sniffing tool used to capture and analyze network traffic.
2. Telnet Client: A Telnet client software installed on a computer to establish Telnet sessions with devices on the network.

Commands:

sudo wireshark

telnet telnet <hostname or IP address> telnet 192.168.1.100

B.3 Question of Curiosity:

1. Which command is need to configure telnet in router?
2. What are the steps needed to extract data from sniffed traffic?
3. What type of packets to be filtered for accessing remote login username and password of a router?
4. What are the differences between Telnet and SSH in terms of security?
5. What filtering techniques can be used in Wireshark to analyze Telnet sessions?
6. How can encrypted protocols mitigate the risks identified in sniffed Telnet traffic?

CSS Expt 6

Q1) Which command is needed to configure Telnet access?

→ To configure Telnet on a router, you need to enable the Vty (Virtual Teletype) lines & set a password for remote access.

1. Enter global configuration mode:
Router > enable

Router # configure terminal

2. Access the vty line configuration mode.
Router (config) # line vty 0 4

3. Set the password for telnet access.

Router (config-line) # password [your-password]

4. Enable login to use the password.

Router (config-line) # login

5. Exit the line config & return global

Router (config-line) # exit

7. Enable telnet & allow remote login access.

Router (config) # transport input telnet

8. Exit configuration mode & save changes.

Router (config) # exit

Router # write memory

2) What are the steps needed to extract data snuffed traffic?

→ 1. Capture the traffic.

- Use a packet sniffer like Wireshark, tcpdump to capture network traffic on the target network interface.

2. Follow traffic based protocol.

- If you're looking for specific data like HTTP, filter the traffic protocol to narrow down captured packets.

3. inspect packet contents

- Once the traffic is captured, examine the packet contents to extract useful infoⁿ
- look for plaintext data or protocols that do not encrypt traffic

4. Extract appⁿ data

- For telnet, you may find username, password if not encrypted
- For DNS, review DNS queries & response
- For FTP, you may extract username & passwords

5. Analyze & filter sensitive data

- Use additional tools like grep, awk or sed to customize extraction & parsing of the relevant data.

3) ⇒ 1. Telnet:

- Telnet sends data in plaintext (unencrypted), To capture username, password during Telnet session, on port 23.
- Filter for TCP port 23 in your packet sniffer (e.g. Wireshark)
- Look for login prompts & username/password pairs within the captured Telnet traffic.

tcp.port == 23.

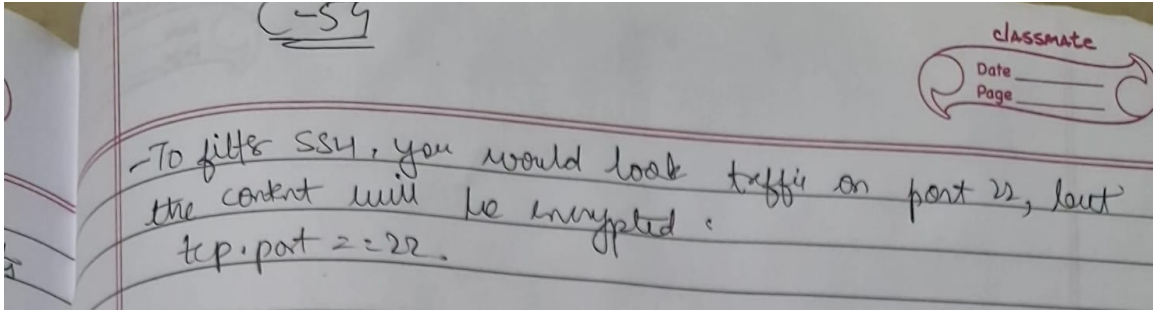
2. FTP:

- If the router is accessed via FTP which can also transmit credentials in text, filter for FTP traffic on port 21.
- Look for user & pass commands in the FTP control stream.

if tcp.port == 21

3. SSH (Secure Shell):

- SSH encrypts all traffic, including login credentials. As a result, sniffing SSH traffic for usernames & passwords is feasible without breaking the encryption.



B.4 Conclusion:

(Write appropriate conclusion.)

In conclusion, designing a network and implementing packet sniffing on Telnet traffic using Wireshark requires careful planning, ethical considerations, and a focus on security. By analyzing captured packets and identifying potential vulnerabilities, organizations can improve their network security posture and mitigate risks associated with insecure protocols like Telnet.